

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
 In preparation for General Certificate of Education Advanced Level
 Higher 2

CANDIDATE
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BIOLOGY

9648/03

Application Paper and Planning Question

19 September 2016

Paper 3

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

All working for numerical answers must be shown.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question

For Examiner's Use	
1	
2	
3	
4 Planning	
5	
Total	

This document consists of **17** printed pages including the cover page and **1** blank page.

[Turn Over

Answer **all** questions.

- 1 Bloom's syndrome is a rare autosomal recessive disorder. It is characterized by short stature, predisposition to the development of cancer and genomic instability. Bloom's syndrome is caused by mutations in the BLM gene leading to mutated DNA helicase protein formation.

In the early 1980s, attempts were made to map the BLM gene on chromosome 15 using DNA markers. The relationship between the BLM gene and two DNA markers DM1 and DM2 were studied in a family with many affected children. DNA were extracted from these individuals and cut with two different restriction enzymes separately. The products of digestion were separated using gel electrophoresis and the RFLP banding patterns is shown in Fig. 1.1 below.

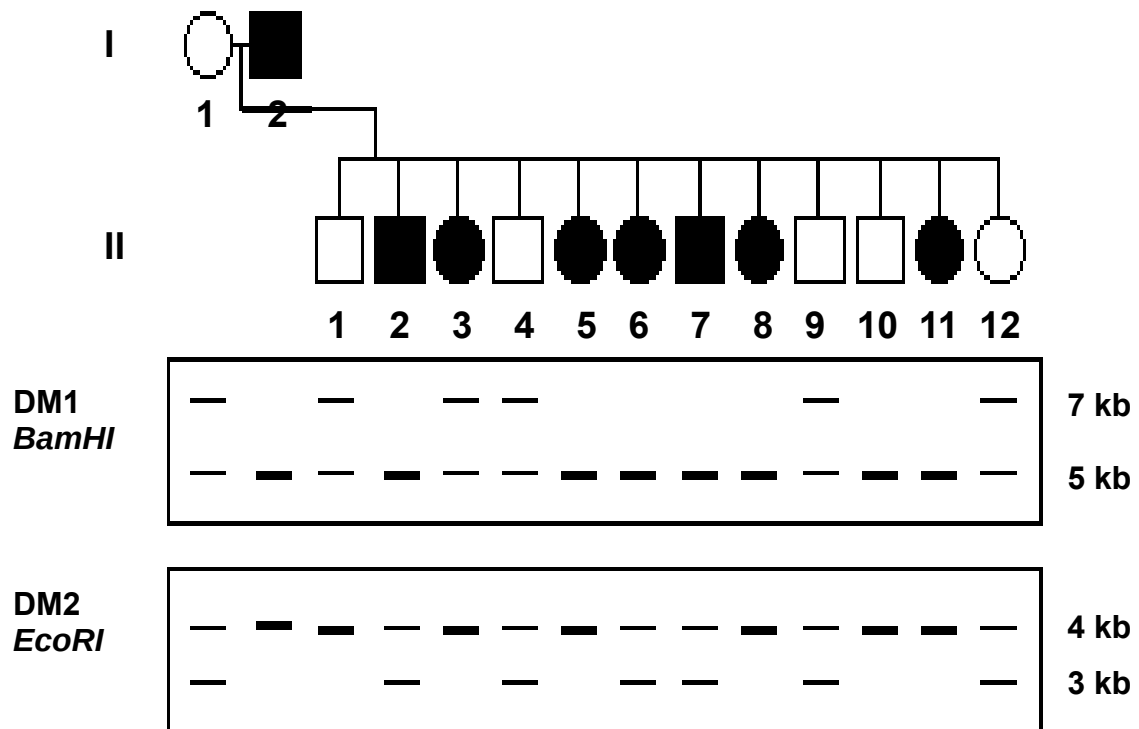


Fig. 1.1

(Note: *Bam*HI was used on DM1 and *Eco*RI on DM2)

Key

	Male with Bloom's syndrome		Female with Bloom's syndrome
	Normal male		Normal Female

- (a) Explain why genes are seldom used as markers for genetic testing. [2]

- a) genes encodes a functional polypeptide;
- b) sequence of DNA is normally conserved due to its important biological function;
- c) mutation rate in genes is lower than in non-coding regions;
- d) not able to differentiate between individuals;

- (b) What is meant by a restriction enzyme? [2]

- a) recognise a specific sequence at the RE site;
- b) ref. RE site being 4-6 bases long;

- c) and binds to the specific sequence of bases;
 - d) cuts/cleaves DNA;
 - e) through hydrolysis of phosphodiester bonds between nucleotides;
- Max 2

(c) Explain the principles behind Restriction Fragment Length Polymorphism (RFLP) analysis. [3]

- a) different individuals have genetic variations / different nucleotide sequences at same locus;
 - b) among individuals of a species;
 - c) caused by mutations;
 - d) found within coding or non-coding region;
 - e) giving rise to Gain or loss restriction sites;
 - f) cut with same RE;
 - g) result in different length of DNA fragments;
- Max 3m

(d) Explain how gel electrophoresis can be used to distinguish between the different band patterns as observed in Fig. 1.1 using *Bam*HI. [3]

- a) DNA fragments can be separated based on molecular size using gel electrophoresis;
 - b) The gel provides a matrix, in which the DNA fragments have to maneuver through the pores of the gel;
 - c) Thus, in a fixed amount of time;
 - d) the larger DNA (7kb) will move a shorter distance, compared to a small fragment(5kb), which will move a longer distance thus nearer to the well;
 - e) DNA is negatively charged at neutral pH due to the phosphate group;
 - f) The wells in the gel are located near the cathode;
 - g) therefore DNA sample will migrate from negative (cathode) to positive (anode) poles when subjected to an electric field;
- Max 3m

(e) Using the information provided in Fig. 1.1, deduce and explain which marker is more suitable in detecting the presence of Bloom's syndrome. [2]

- a) DM1 – more suitable OR DM2 – less suitable;;

Either

- b) For DM1, 7 out of 8 (88%) of the individuals with BS have the only one 5kb band/ homozygous for the 5 kb fragments;
- c) 5 out of 8 (62.5%) of the BS individuals have only one 4kb band/ homozygous for the 4 kb fragments for DM2;

Or

- d) For DM1 - All individuals with Bloom's syndrome have only one 5kb band/ homozygous for the 5 kb fragments except II-3 who has 2 bands 5kb and 7 kb;
- e) For DM2 - 5 BS individuals have only one 4kb band/ homozygous for the 4 kb fragments and 4 BS individuals have 2 bands 3kb and 4kb / about equal number of BS individuals and normal individuals have a single 4kb band as well as 2 bands 3kb and 4kb;
- f) (So the inheritance of the DM1/5kb fragment is more consistent compared to DM2)

- (f) Using your understanding of linked genes, provide a possible explanation for your answer in (e). [2]

- a) DM1 closer to BLM gene compared to DM2;;
(note: some form of comparison required between DM1 and DM2 before a full mark is given)
- b) Higher tendency for the 5kb fragment to be inherited together with the BLM gene;
- c) Hence lower number of recombinants for DM1 (individuals II-3 and II-10) compared to DM2 (II-1,2,6,7 and 10);

[Total: 14]

- 2 Severe Combined Immunodeficiency Disease (SCID) is the most severe of the immune deficiency diseases. SCID is a group of very rare, life-threatening diseases that are present at birth. Fig. 2.1 shows the pattern of inheritance of X-linked SCID in a family.

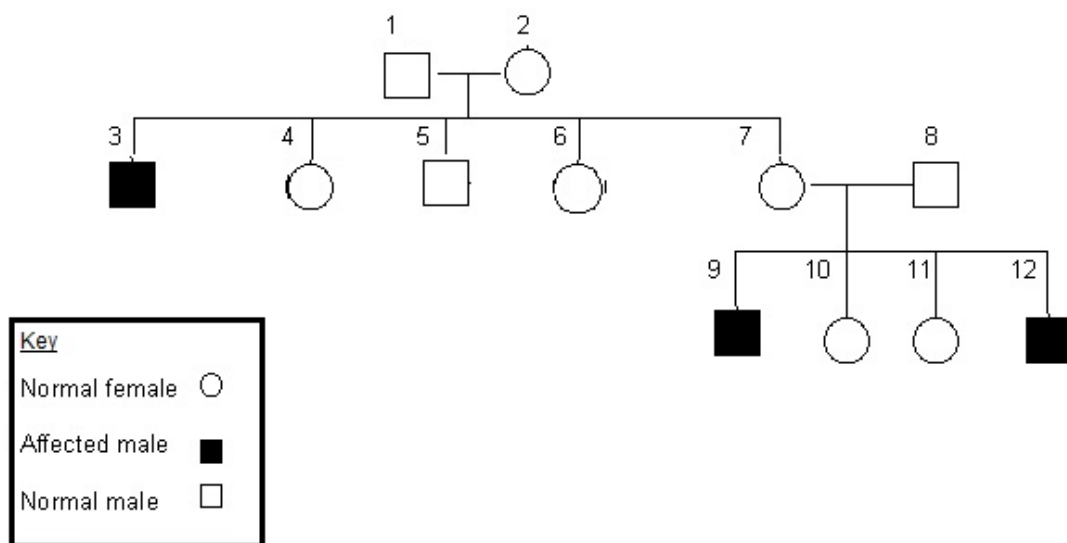


Fig. 2.1

- (a) State and explain the nature of alleles that gives rise to SCID illustrated in Fig. 2.1. [2]

- a) Recessive alleles;;
- b) Unaffected parents 1 & 2 have affected son 3/ Unaffected parents 7 & 8 with affected sons 9 & 12;;

- (b) Describe the genetic basis of X-linked SCID. [3]

- a) Interleukin-2-Receptor Gamma (IL2RG) gene found on the X chromosome;
- b) encodes common gamma chain subunit of the interleukin receptor;
- c) These receptors reside in the plasma membrane of immune cells;
- d) and allow communications between T and B cells/ immune cells;
- e) Mutations of IL2RG gene results in a non-functional version of the common gamma chain, or no protein formation;
- f) Without the common gamma chain, these interleukin-2-receptors cannot form /are absent from immune cells;
- g) hence preventing communication between T and B lymphocytes;
- h) These interleukins and their receptors are also involved in the development and differentiation of T and B cells;

- i) The result is a near complete failure of the immune system to develop and function / low number or absence of T cells and non-functional B cells (B cells need T-helper cells to function);

Max 3m

(c) Give one reason why SCID is a suitable choice for gene therapy. [1]

- a) A mutation in 2 copies of IL2RG gene/ADA gene gives rise to SCID;
b) Only require one copy of normal/functional allele to be delivered to target cell for restoring the normal phenotype;

Babies born with SCID have no defence against common infections and quickly become ill when the protection from maternal antibodies is lost. Gene therapy for SCID has been carried out using the general procedure shown in Fig. 2.2.

Step 1: Hematopoietic stem cells retrieved from baby's umbilical cord blood.



Step 2: Stem cells infected with harmless genetically engineered virus containing the normal, dominant allele.



Step 3: Stem cells take up normal allele.



Step 4: Stem cells transfused back into baby.



Step 5: Immune system develops T and B lymphocytes.

Fig. 2.2

(d) Using the information given, state **two** reasons why stem cells were used in this treatment. [2]

- a) multipotent + differentiate into limited range of cell type;;
b) Unspecialised, no tissue-specific structures that allow specialized functions;;
Reject: undifferentiated only
c) capable of dividing by mitosis and renew for long period of time/long term proliferation;;
d) stem cells used belong to the patient's hence there is low/no risk of immune rejection;;

In 2002, two young patients with X-linked SCID were treated with their own blood stem cells that had been subjected to the same treatment as the earlier attempts at gene therapy.

The two different forms of SCID require different cells to be targeted for gene therapy. For X-linked SCID, only stem cells can be used while in autosomal recessive form of SCID (ADA), both T cells and stem cells could be used. If T cells are used, they will be induced to proliferate *in vitro* by treatment with relevant signalling molecules before infection with viral vectors.

(e) From the information given and your knowledge of SCID, explain why different cells are used for different treatments. [2]

- a) In X-linked SCID, IL2RG gene is mutated which results in the absence of functioning interleukin receptors;
- b) The patient's T cells cannot be stimulated to proliferate by treating them with Interleukin 2 (IL-2) before the infection with viral vector; therefore, T cells are not used for gene therapy for X-linked SCID and stem cells are used instead
- c) In autosomal recessive form of SCID, normal IL2RG gene is present;
- d) interleukin 2 receptors are functional in immune cells and can bind to IL2 therefore, both T cells and stem cells can be targeted;

Fig. 2.3 shows the lymphocyte counts (both T & B) of two infant patients that underwent two separate types of gene therapy treatments for SCID. Patient A was treated with a retrovirus while Patient B was treated with an adenovirus. The lymphocyte count of the patients was monitored by a doctor at the end of every month for a period of 12 months. (The acceptable range of lymphocyte count for non-SCID patients is 5 to $10 \times 10^3/\text{mm}^3$).

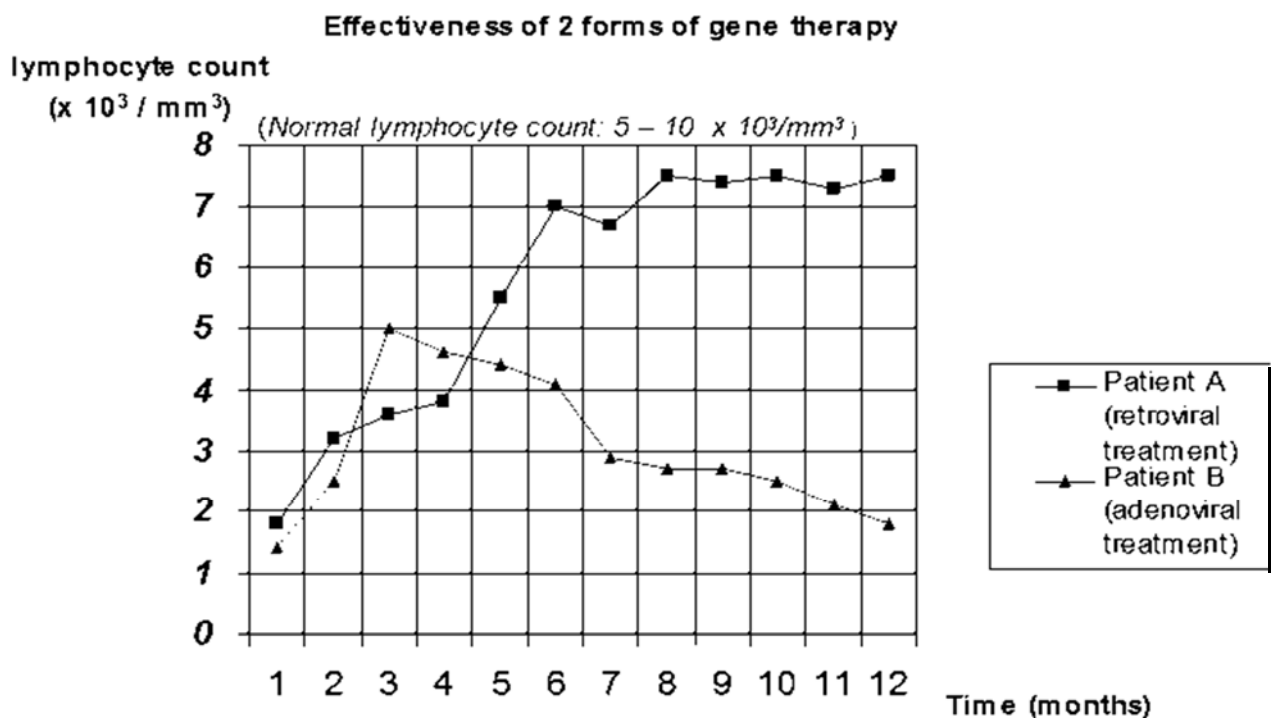


Fig. 2.3

(f) With reference to Fig. 2.3,

(i) describe the trends observed in both patients' lymphocyte count. [2]

- a) Patient A lymphocyte count increase from 1.8 to $7.5 \times 10^3 / \text{mm}^3$ from 1 to 12 months / entered normal range by 5th month;;
- b) Patient B lymphocyte count increase to $5 \times 10^3 / \text{mm}^3$, from 1 to 3 months but decreased thereafter to $1.8 \times 10^3 / \text{mm}^3$ from 3 to 12 months / entered into normal range by 3rd month;;

(ii) account for the difference in lymphocyte count at the end of 12 months. [2]

- a) retroviral vector allowed for integration of recombinant DNA that could be stably propagated hence resulting in long term stable expression / pass from parent to daughter cells;;
- b) adenovirus however could not integrate / exists extrachromosomally, into chromosomal DNA hence transferred gene does not segregate equally into daughter cells;;

Reject: DNA gets degraded

[Total: 14]

- 3 In 2012, permission was granted for a field trial in the UK of genetically modified wheat, *Triticum aestivum*. The wheat carries a gene, taken from peppermint plants, which results in the wheat leaves releasing a volatile, non-toxic chemical, (E)- β -farnesene (E β f), into the atmosphere.

E β f is not only produced by various species of plants. It is also secreted by aphids when they are disturbed by a predator.

Two experiments have been performed into the effect of E β f on the behaviour of aphids feeding on leaves in closed containers.

Experiment 1

Either 10 cm³ of air from a syringe that contained plant leaves that secrete E β f
or 10 cm³ of air from a syringe with normal leaves
was added to the containers of feeding aphids.

Experiment 2

Either 10 cm³ of air containing 50 ng of E β f
or 10 cm³ of air containing no E β f
was added to the containers of feeding aphids.

In both experiments, the number of aphids that stopped feeding and moved away from the food leaves was counted. The results are shown in Table 3.1.

Table 3.1

	Experiment 1		Experiment 2	
Air added to containers of feeding aphids	10 cm ³ air that had been in contact with leaves secreting E β f	10 cm ³ air that had not been in contact with leaves secreting E β f	10 cm ³ air containing 50 ng E β f	10 cm ³ air containing no E β f
Number of aphids in container	99	113	132	106
Number of aphids that stopped feeding and moved away from the food leaves	54	1	111	0

- (a) With reference to Table 3.1, describe the evidences which support the idea that E β f is an alarm signal for aphids. [2]

Either

- a) In the presence of E β f, large number of aphids stopped feeding and moved away from food source (leaves), with 55% / 54 out of 99 aphids in experiment 1 compared to 84% / 111 out of 132 aphids in experiment 2 respectively;;
- b) In the absence of E β f, few / no aphids stopped feeding and moved away from food source with 1 aphid and none, in experiment 1 and 2 respectively;;

Or

- c) In experiment 1, 55% / 54 out of 99 aphids that were exposed to air that had been in contact with leaves secreting E β f, stopped feeding and moved away from food source, compared to 0.9% / 1 out of 113 aphids that were exposed to air that had not been in contact with leaves secreting E β f, stopped feeding and moved away from food source;;
 - d) In experiment 2, 84% / 111 out of 132 aphids that were exposed to air containing 50ng E β f, stopped feeding and moved away from food source, compared to 0% / none of the 106 aphids that were exposed to 20 cm³ air containing no E β f, stopped feeding and moved away from food source;;
- (b) Other experiments show that E β f attracts predators of aphids, such as ladybirds. Suggest how growing genetically modified wheat secreting E β f could increase the yield of wheat. [1]

Either

- a) E β f attracts predators of aphids / ladybirds which will prey on aphids;

Or

- b) attacked aphids / aphids disturbed by predators / ladybirds will secrete even more E β f, which attracts more predators / ladybirds;
 - c) Aphids are not able to feed on wheat / obtain nutrients from GM wheat;
- (c) Suggest why growing this genetically modified wheat might be acceptable to people who object to the growth of genetically modified insect-resistant maize or cotton. [2]
- a) gene encoding for E β f is already found in peppermint / various plant species' genomes;;
 - b) E β f is not toxic / harmful to human health / no new chemical is added to human diet;;
 - c) Unlike Bt maize or Bt cotton, E β f does not kill insects (since it is naturally secreted from aphids);;
 - d) Aphids are thus still available for their predators / minimal impact on the food chain;;

In a separate experiment, Ti (Tumour-inducing) plasmid found in *Agrobacterium tumefaciens* was used to transfer DNA of interest into *T. triticum*. The transformed wheat protoplasts are stimulated to divide by mitosis to form calli and regenerated into whole plants by tissue culture techniques.

- (d) With reference to tissue culture techniques, describe how plant growth regulators can result in plantlet formation from calli. [2]

(Achieved by varying the ratio of plant growth regulators)

- a) Higher proportion of auxin to cytokinin stimulates root growth and cell elongation;;
- b) Higher proportion of cytokinin to auxin stimulates shoot growth and cell division;;

Plasmids are small circles of DNA that can be found in many bacteria which can be used for cloning foreign genes in suitable bacteria.

The plasmid, pUC19, shown in Fig. 3.2 has been developed for specific purposes by genetic engineers.

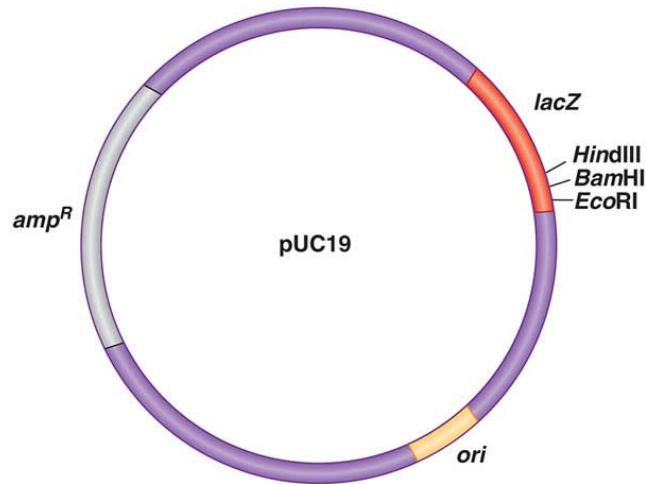


Fig. 3.2

pUC19 can be cleaved using restriction enzymes such as HindIII. HindIII can be found naturally in *Haemophilus influenzae*.

(e) What is the natural role of restriction enzymes in *H. influenzae*? [2]

- a) Protect bacterium against invading virus DNA/ foreign DNA;
- b) by cutting it up at specific sites (via breaking of phosphodiester bonds);
- c) Thus the enzyme helps to restrict foreign DNA from surviving/replicating in bacterium;
- d) Bacterium's own DNA is protected from digestion by methylation to the restriction sites;

R: any usefulness to humans in genetic engineering

(f) With reference to Fig. 3.2, explain how the cells which contain the recombinant plasmids are identified. [3]

- a) The bacteria are plated on an agar plate containing ampicillin to eliminate those with no uptake of the cloning vectors/untransformed cells (OWTTE);
- b) As those cells that are not transformed do not have the ampicillin resistant gene are not able to produce the enzyme that breaks down ampicillin;
- c) The same agar plate also contains X-Gal to select for cells with recombinant plasmids;
- d) Clones of bacteria that contain the gene of interest will appear white while those that do not contain the gene of interest will appear blue;
- e) As the gene of interest caused the insertional inactivation of the lac Z gene;
- f) β -galactosidase is not produced to break down X-gal, hence the colony remains white;
- g) Cells with normal plasmids will have an intact lac Z gene;
- h) which can be transcribed to produce β -galactosidase to break down X-gal to a blue product, causing the colony to appear blue;

Max 3m

[Total: 12]

4 Planning question

Dormant seeds have a very low rate of respiration. When water is absorbed by dormant seeds, growth hormones are activated. These hormones activate genes that code for enzymes that hydrolyse stored food reserves used in respiration allowing the seed to grow and germinate. The respiration rate can be measured by oxygen usage per unit mass using a respirometer.

You are required to investigate the respiration rate of germinating seeds of different mass.

You are provided with the following materials and apparatus which you must use:

- Mung bean seeds of different mass that have been soaked in water for 24 hours
- Blue dye
- Soda lime pellets 
- Glass beads
- Rubber tubing connected to capillary tube of 4mm diameter
- Syringes
- Electronic weighing balance
- Scale paper

You may select from the following apparatus and use appropriate additional apparatus:

- Normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- Forceps
- Ruler
- Stopwatch
- Thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Suggested answer

Aim:

To investigate the respiration rate of germinating seeds of different mass.

Background:

- Ref to intake of H_2O , activation of growth hormone & enzyme that hydrolyse food reserves for respiration
- ref to aerobic respiration for production of ATP for cell division / growth
- need for O_2 in respiration as final electron acceptor;;
- relate measurement of O_2 uptake as indication of respiration rate

Null Hypothesis: Mung beans of different mass may have same / different respiration rate with logical suggestion e.g. same spp thus likely to be the same rate;;

Experimental Procedure:

Labelled diagram

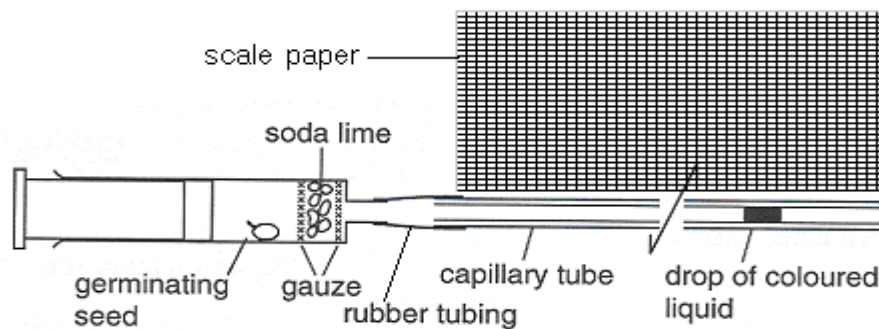


Fig. 1.0

- Set up the experiment in sets of 3.
- Remove the plunger from the syringe and place soda lime granules inside the syringe as shown in the set up above.
- Using a forceps, take 1 mung bean seedling and carefully remove and discard its testa (seed coat).
- Weigh it on the electronic weighing balance. Record your results.
- Place the mung bean in the syringe barrel and replace the plunger by pushing it in until it is about 0.5 cm from the seedling.
- Connect the glass capillary tube securely to the syringe using a rubber connecting tubing.
- Dip the end of the glass capillary tube into the blue dye so that a drop enters the capillary tube. (see the set up above)
- Remove any excess liquid with paper toweling.
- Place the respirometer horizontally on a piece of graph paper. Start the stopwatch and wait for 3 minutes to ensure that the manometer fluid is moving smoothly towards the syringe. Stop the stopwatch.
- Without handling the apparatus, measure the distance travelled by the manometer fluid in 3 consecutive time intervals of 1 min.

11. Do this by marking the position of the fluid on the graph paper and reading off the distances.
(Alternatively, a piece of white paper can be used, and the markings measured using a ruler).
12. Record these results in the following table.

Table of results recording the distance travelled by 5 different mung beans in each minute/mm

Mung Bean	Weight of Mung Bean/g	Minute	Distance travelled in each minute/mm	Average distance travelled in each minute/mm	Rate of Respiration /min ⁻¹ g ⁻¹
1		1			
		2			
		3			
2		1			
		2			
		3			
3		1			
		2			
		3			
4		1			
		2			
		3			
5		1			
		2			
		3			

13. Detach the syringe from the capillary tube by pulling it gently from the rubber connecting tube.
14. Fit another empty syringe to the capillary tube and flush out the manometer fluid onto a piece of filter paper so that the bore of the capillary tube is empty.
15. Carry out the same procedure for the rest of the 4 mung beans.
16. Repeat the entire experiment from step 1 to 15 twice more to ensure reproducibility of the experiment with fresh materials.
17. Calculate the rate of respiration in mm³ of O₂ taken in min⁻¹ g⁻¹

Variables

Independent variable: 5 different masses of mung beans of reasonable range & regular intervals/g, weigh using electronic balance;;

Dependent variable: respiration rate measured by vol of O₂ uptake, inclusion of soda lime to absorb CO₂, calculated from distance moved by dye droplet;;

Control: description of an appropriate setup using glass beads that occupies similar vol as mung beans to ensure that the mung beans are directly involved in the reaction.

Safety Precautions:

1. Soda lime is corrosive and a skin irritant. Be careful when handling this chemical and to be sure goggles and gloves are worn at all times.
2. Handle glass wear with caution to prevent breaking of any fragile items.

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Describe PCR and explain the advantages and limitations of this technique.

[8]

Stage 1: Denaturation

- a) by heating to 90°C; [accept 90 – 100°C]
- b) DNA is separated into single strands by; **R: unwind**
- c) breaking of hydrogen bonds;

Stage 2: Annealing of primers

- d) by cooling to 54°C; [accept 30 – 65°C]
- e) single-stranded;
- f) primers/ oligonucleotides bind to single (DNA) strands / 3' ends;
- g) by annealing/hybridise;
- h) to their complementary sequences on either side of the target sequence;

Stage 3: Extension

- i) by heating to 72°C; [accept 60 – 75°C]
- j) optimum temperature (for Taq polymerase);
- k) new strands (of DNA) are synthesised by Taq/ DNA polymerase;
- l) starts at position of primers;
- m) addition of free deoxyribonucleotide to the free 3'OH end;
- n) through the formation of phosphodiester bond between the nucleotides;
- o) using the single strand DNA (target sequence) as a template;

Points a) to o): ½ m each (max 5 m)

Advantages of PCR:

- p) Fast and efficient way to amplify DNA with the exponential increase of the amount of DNA because the process of PCR can be automated;;
- q) PCR is highly sensitive. A target sequence can be amplified even when there is only a very small amount of the DNA available;;
- r) Give one useful example;;
 - Many copies of a gene of interest can be amplified for cloning purposes, as long as the primer sequences are known.
 - Clinical diagnosis e.g. in prenatal diagnosis of human genetic disorders. DNA from single embryonic cells is amplified using PCR. Screening for certain genetic diseases e.g. cystic fibrosis even before the phenotype is expressed provides assurance to the couple. If child is diagnosed with certain diseases, couple can then be prepared and make informed choices.
 - Early detection of infection with HIV: PCR can detect the presence of the HIV genome at very early stages of the diseases before symptoms appear.

- Forensic science: minute amount of DNA found e.g. at crime scene can be amplified to sufficiently large amounts to be analysed.
- Study of fossil: PCR is used to amplify fragments of ancient DNA from a 40,000 year old frozen woolly mammoth.

Limitations of PCR:

- s) Taq polymerase lacks the 3' to 5' proofreading mechanism. This makes it impossible for the enzyme to check the base it has just inserted and remove it if it is incorrect. Polymerase induced errors may accumulate during PCR;;
- t) PCR requires at least the knowledge of the sequences that flank the target DNA to synthesis the primers. Without proper primers, then PCR cannot take place;;
- u) DNA fragments are limited to 3 kb. Further increase in length will decrease the efficiency of amplification;;

Points p) to u): 1 m each

Max 2m each for advantages/limitations

- (b) Describe how plant tissue culture is used to clone plant cells. Explain the scientific reasons for each step in the process. [8]
- a) In micropropagation, plant tissues or explants e.g. meristematic cells are removed from plants or explants only;
- b) Reason: Meristematic regions are preferred as they are undifferentiated and virus-free;
- c) Surface of explants sterilised with dilute sodium hypochlorite (Clorox);
- d) Reason: To kill bacterial and fungal pathogens or organisms;
- e) And grown in a containing sterile media containing nutrients and plant growth regulators needed for plant growth;
- f) Containers holding the explants are then sealed and incubated for 1-9 weeks;
- g) Reason: All procedures must be sterile to ensure that microorganisms do not grow in the cultures as they would grow faster and out-compete the plants in these conditions;
- h) During this period, the cultured cells divide by mitosis;
- i) to form a mass of undifferentiated tissue called a callus;
- j) As callus increases in size, pieces of callus is sliced off and grown on new medium composition (subculturing);
- k) Reason: This creates many genetically identical clones of the original plant;
- l) By adjusting concentration of auxin to cytokinin ratio in growth medium;
- m) cells in callus can be induced to differentiate into roots and shoots;
- n) Reason: To enable plantlet formation;
- o) Further growth being encouraged by the use of plant growth substances until plantlets are large enough;
- p) Plantlets are taken out from culture vessels, washed to remove the agar, soaked in fungicide and antibiotics and then planted in sterile soil in green house for 4-8 weeks;
- q) Reason: This is to allow the plants to acclimatise from a heterotrophic to an autotrophic existence and the acclimatization of the plant to the outdoor environment;
- r) After acclimatization in green house, the plant are transferred to soil for field planting;

Points a) to r): ½ m each (max 8 m)

(c) Distinguish between a genomic DNA library and a cDNA library.

[4]

Genomic DNA library	cDNA library
a) Contains all coding and non-coding sequences;	Expressed genes in the tissue – lack non-coding regions;
b) Cell's DNA cut using RE and inserted into vectors;	mRNA → cDNA using reverse transcriptase which is then inserted into vectors;
c) Need vectors ranging from plasmids to BAC to accommodate larger DNA sequences;	cDNA is relatively short, plasmids and λ phage often used as vectors;
d) A single gene can be spread out over different clones - A lot of DNA to isolate and analyse;	only contain coding sequences for that gene, which reduces the amount of DNA one has to isolate and analyse;
e) To study control of gene expression;	Isolate a particular gene that is active in a specialized cell and analyse its properties;

Any 4 differences, must be point to point comparisons (max 4 m)

[Total: 20]